Project Proposal

On

E-Commerce Management System

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Created By:

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1. Title of the Project

E - COMMERCE MANAGEMENT SYSTEM

2. Introduction

In the modern digital era, online shopping has become an essential part of daily life. Customers prefer buying products online due to convenience, time-saving, and easy comparison of products. An **Ecommerce System** provides a platform where customers can browse products, place orders, make payments, and give reviews through the internet.

This project, **ecommercesystem**, is a database-driven online shopping management system. It maintains customer information, product details, categories, orders, payments, cart items, and reviews. The system is designed to support both customers and administrators. Customers can register, login, add products to cart, place orders, and make payments. The administrator manages products, categories, and monitors orders.

The backend database is developed using **MySQL**, ensuring data consistency, integrity, and easy retrieval of information.

3. Objective

The main objectives of the Ecommerce System are:

- To provide an online platform for buying and selling products.
- To store customer, product, order, and payment details securely.
- To allow customers to browse products by categories.
- To manage shopping cart and order placement.
- To maintain transaction records for payments.
- To allow customers to give reviews and ratings.
- To help administrators manage products and categories.
- To reduce manual work and paperwork.
- To provide fast, reliable, and accurate data handling.

4. PROJECT CATEGORY

Database and Java Application Based Project

Type: **Command-Line Based Application**

5. Analysis

5.1 Modules and Description

1. Customer Module

- Customer registration and login.
- Stores personal details (name, email, phone, address).
- Customer can add products to cart.
- Customer places orders and makes payments.
- Customer gives product reviews.

2. Admin Module

- Admin login.
- Add, update, delete product categories.
- Add, update, delete products.
- Monitor customer orders and payments.

3. Product Module

- Stores product details: name, description, price, stock.
- Each product belongs to a category.

4. Category Module

- Stores product categories (Electronics, Clothing, Books, etc.).

5. Cart Module

- Stores temporary products selected by customers before ordering.

6. Order Module

- Stores order details: total amount, status, shipping address, payment method.

7. Payment Module

- Stores payment details of each order.

8. Review Module

- Stores customer feedback and ratings for products.

5.2 Database Design

The database name is **ecommerce_db**.
It contains the following tables:

Table 1: Admin_Users

Field	Type	Null	Key	Default	Extra
admin_id	int	NO	PRI	NULL	auto_increment
username	varchar(50)	NO	UNI	NULL	
password	varchar(100)	NO		NULL	
name	varchar(100)	NO		NULL	

Table 2: Cart_Items

Field	Type	Null	Key	Default	Extra
cart_id	int	NO	PRI	NULL	auto_increment
customer_id	int	NO	MUL	NULL	
product_id	int	NO	MUL	NULL	
quantity	int	YES		1	
added_date	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

Table 3 : Categories

Field	Type	Null	Key	Default	Extra
category_id	int	NO	PRI	NULL	auto_increment
name	varchar(100)	NO		NULL	
description	text	YES		NULL	

Table 4: Customers

Field	Type	Null	Key	Default	Extra
customer_id	int	NO	PRI	NULL	auto_increment
name	varchar(100)	NO		NULL	
email	varchar(100)	NO	UNI	NULL	
password	varchar(100)	NO		NULL	
phone	varchar(15)	YES		NULL	
address	text	YES		NULL	
created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

Table 5: Order_items

Field	Type	Null	Key	Default	Extra
order_item_id	int	NO	PRI	NULL	auto_increment
order_id	int	NO	MUL	NULL	
product_id	int	NO	MUL	NULL	
quantity	int	NO		NULL	
price	decimal(10,2)	NO		NULL	

Table 6: Orders

Field	Type	Null	Key	Default	Extra
order_id	int	NO	PRI	NULL	auto_increment
customer_id	int	NO	MUL	NULL	
order_date	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
total_amount	decimal(10,2)	NO		NULL	
status	varchar(20)	YES		PENDING	
shipping_address	text	YES		NULL	
payment_method	varchar(20)	YES		NULL	

Table 7: Payments

Field	Type	Null	Key	Default	Extra
payment_id	int	NO	PRI	NULL	auto_increment
order_id	int	NO	MUL	NULL	
amount	decimal(10,2)	NO		NULL	
payment_date	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED
method	varchar(20)	YES		NULL	
status	varchar(20)	YES		PENDING	

Table 8: Products

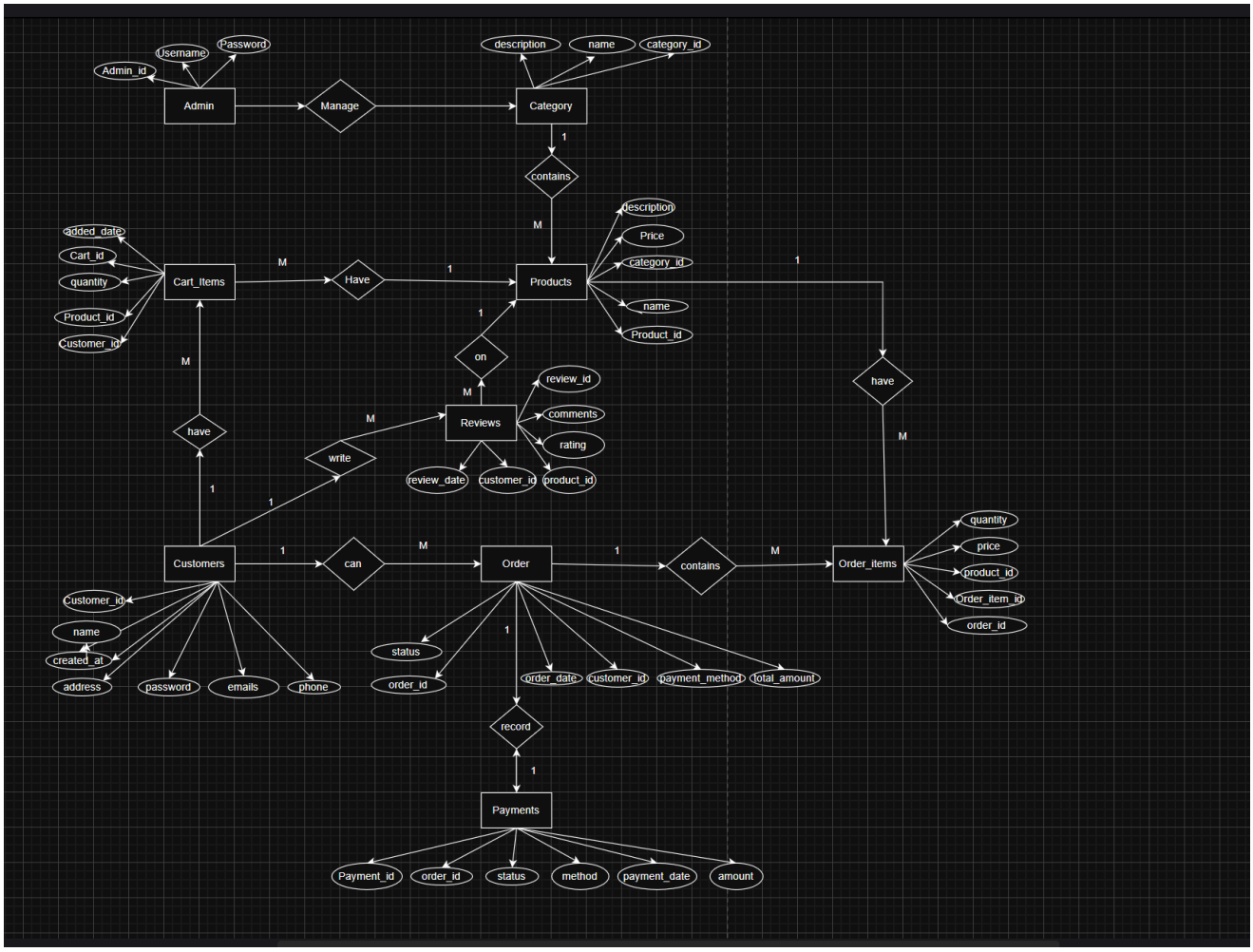
Field	Type	Null	Key	Default	Extra
product_id	int	NO	PRI	NULL	auto_increment
name	varchar(200)	NO		NULL	
description	text	YES		NULL	
price	decimal(10,2)	NO		NULL	
stock	int	YES		0	
category_id	int	YES	MUL	NULL	
created_at	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

Table 9: Reviews

Field	Type	Null	Key	Default	Extra
review_id	int	NO	PRI	NULL	auto_increment
product_id	int	NO	MUL	NULL	
customer_id	int	NO	MUL	NULL	
rating	int	YES		NULL	
comment	text	YES		NULL	
review_date	timestamp	YES		CURRENT_TIMESTAMP	DEFAULT_GENERATED

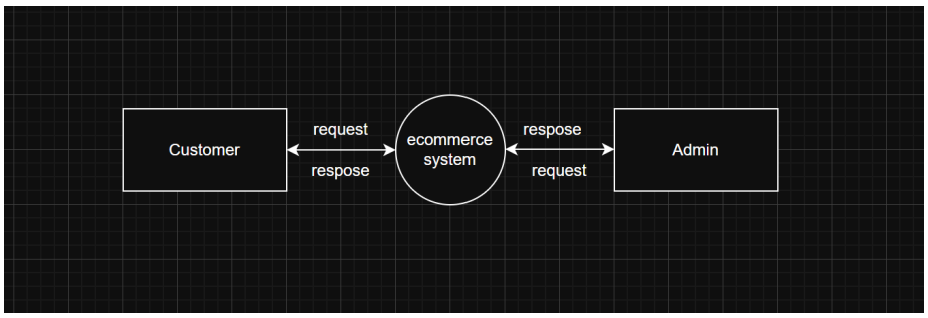
5.3 ER Diagram

Entities and Attributes

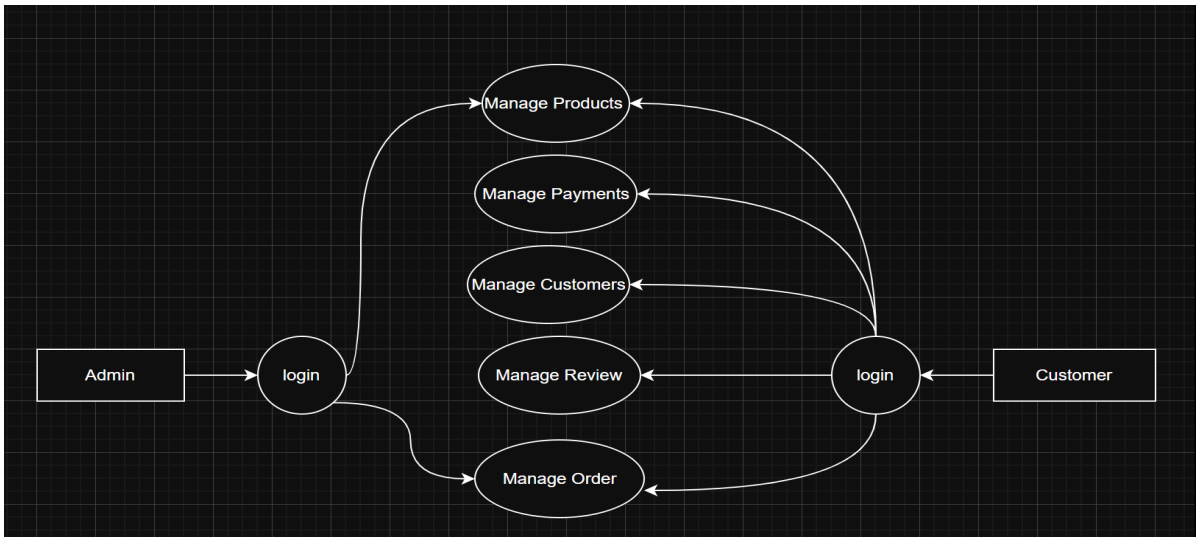


5.4 Data Flow Diagram (DFD)

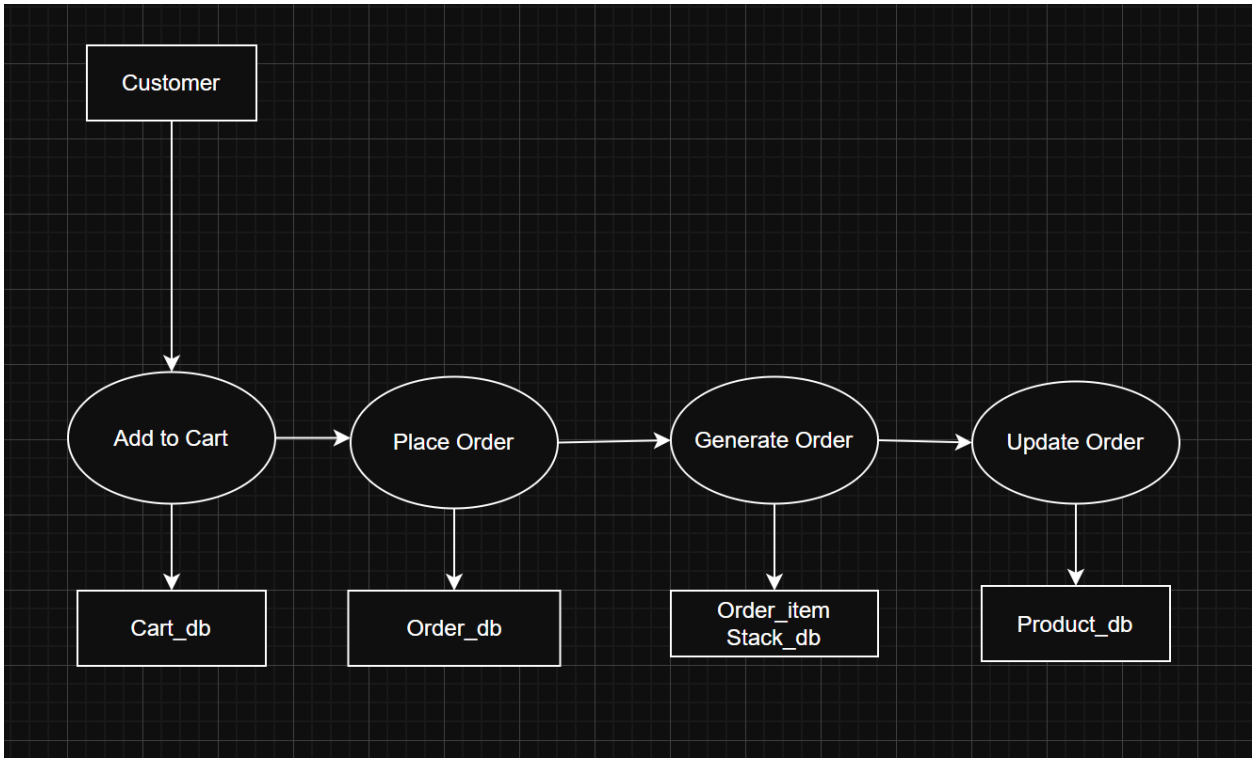
DFD Level 0 (Context Diagram)



DFD Level 1 :

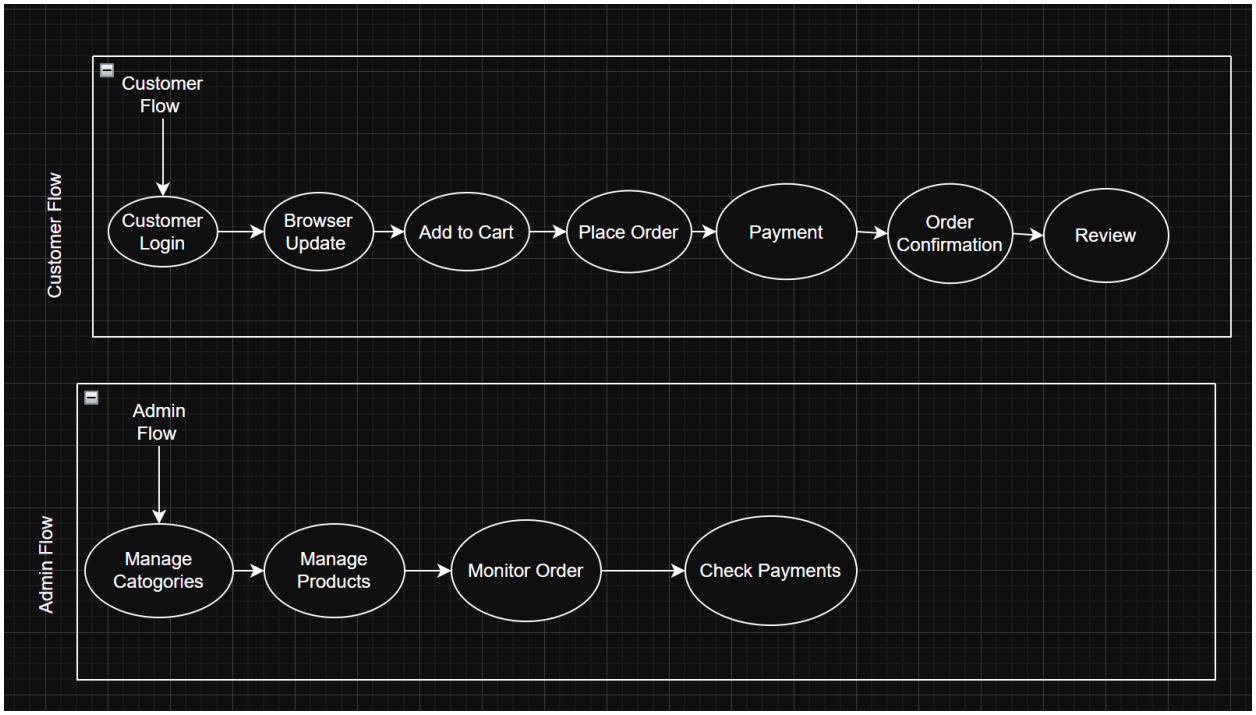


DFD Level 2 :


Customer
Add to Cart
Place Order
Generate Order
Update Order
Cart_db
Order_db
Order_item
Stack_db
Product_db

6. Complete Structure

Process Logical Diagram :



7. PLATFORM USED

Hardware Requirement

- Processor : Intel i3 or above
- RAM : Minimum 4 GB
- Hard Disk : 500 GB
- Keyboard : Standard
- Mouse : Optical

Software Requirement

- Operating System : Windows 10 / 11
- Programming Language : Java (JDK 8 or above)
- IDE : Eclipse
- Build Tool : Maven
- Database : MySQL
- Connectivity : JDBC

8. FUTURE SCOPE

- Convert command-line system to web-based application.
- Add graphical user interface (GUI).
- Integrate online payment gateway.
- Implement product search and filtering.
- Add OTP and email verification.
- Improve security using encryption.
- Add delivery tracking system.

9. BIBLIOGRAPHY

- Java Official Documentation
- MySQL Official Documentation
- JDBC Tutorial (Oracle Docs)
- Maven Documentation
- Software Engineering Notes